

Megatron mg20 Replication

Every RL-step number in the project was re-measured today on the restored Megatron TP1 trainer, after the discovery that the previous replication had silently trained with FSDP2. This is what the scheduler buys end-to-end at batch 32 and a 16k response cap, with error bars, and why the number is where it is.

1.14x

9B step speedup, live EMA scheduler vs BF16 on vanilla vLLM (2 seeds)

1.02x

9B step speedup, uniform INT4 on vanilla vLLM (+28 % tokens)

0 %

token inflation under the scheduler (reward 0.798 vs 0.816)

R² 0.999

training-side time vs tokens on Megatron (was 0.93 on FSDP2)

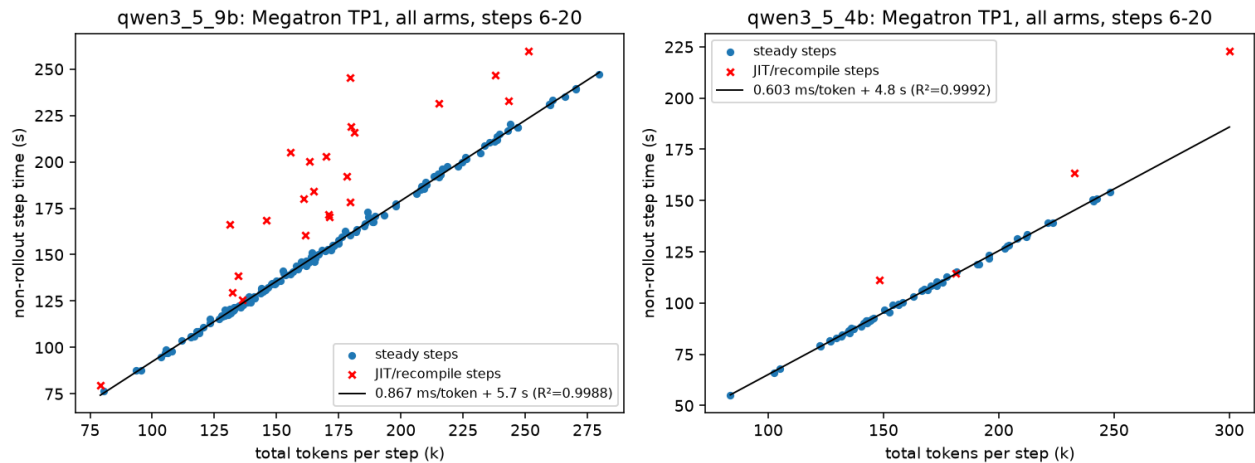
Setup

- **Trainer.** `run_fullstep.sh TRAINER=megatron` : Megatron-Core 0.18 through Megatron-Bridge 0.5, LoRA rank 16 / alpha 16 on the 12 Qwen3.5 projections via `model.lora.*`, TP = PP = 1, full uniform recompute. One A100-80GB per arm, vLLM at `gpu_memory_utilization 0.5`, actor resident (no offload).
- **Workload.** GSM8K, B = 8 prompts × 4 samples = 32 requests per step, response cap 16384, prompt cap 2048, temperature 1, 20 steps. `data.shuffle=False`, so every arm sees the same prompts at each step; rollout seed 42, with a second seed 43 for the headline arms. Steps 1–5 are treated as warm-up and excluded.
- **Baselines.** BF16 and uniform INT4 (W4) on the vanilla vLLM LoRA backend (Punica). Scheduler arms use our fused LoRA fast path + dual stream. So "ours vs BF16" is the whole system: kernels plus precision scheduling.
- **Scheduler arms.** Live EMA (alpha 0.2, calibrated from 128 paired BF16/W4 rollout-only requests and the 9B/4B full-grid heatmaps) at the fitted downstream slope (8.78e-4 s/token, the archived Megatron fit); the same with a 2e-4 s/token INT4 quality penalty; slope 0 as the rollout-only reference; fixed K = 5000.
- **Not included.** Phi-4-mini has no Megatron-Bridge mapping and cannot train on Megatron.

Trainer acceptance

The acceptance criterion was that non-rollout step time is linear in sampled tokens with R² near 1. It is, and the per-token costs are the same in every arm: `old_log_prob` 0.183 ms/token, ref 0.172, `update_actor` 0.505 on 9B.

MODEL	ARMS	STEPS	SLOPE	INTERCEPT	R ² ALL STEPS	R ² STEADY STEPS
Qwen3.5-9B	11	165	8.67e-4 s/tok	5.7 s	0.884	0.9988 (max resid 5.7 s)
Qwen3.5-4B	4	60	6.04e-4 s/tok	4.8 s	0.965	0.9992 (max resid 1.4 s)



Non-rollout step time against total tokens, steps 6–20 of every arm. The red crosses are 27 of 225 steps where the LoRA forward and backward (never the reference pass) ran 20–80 % slower; they cluster at steps 6–8 and at steps that set a new maximum token count, i.e. shape-triggered recompiles. They are kept in every step number below. The 9B slope matches the archived Megatron fit (8.78e-4); the intercept is 5.7 s against 111 s under the offloaded FSDP2 actor.

Results: Qwen3.5-9B

Speedup = BF16-Punica mean time ÷ arm mean time over steps 6–20 (both seeds pooled where two exist). Rollout is `timing_s/gen` ; downstream is everything else in the step.

ARM	RUNS	ROLLOUT	DOWNSTREAM	STEP	TOKENS VS BF16	REWARD	CAP HITS / STEP	SWITCH (TOKENS)
BF16, Punica (baseline)	2	1.00x (246 s)	1.00x (142 s)	1.00x (388 s)	0 %	0.816 ± .018	2.3	—
uniform W4, Punica	2	1.24x	0.79x	1.02x	+28 %	0.742 ± .015	5.5	—
uniform W4, ours	1	1.33x	0.81x	1.08x	+24 %	0.750	5.1	—
fixed K = 5000, ours	2	1.28x	0.92x	1.12x	+7 %	0.800 ± .018	3.4	5000
live EMA, slope 0, ours	1	1.34x	0.85x	1.10x	+15 %	0.777	4.5	1000–2250
live EMA, fitted slope, ours	2	1.27x	0.98x	1.14x (339 ± 12 s)	0 %	0.798 ± .019	2.5	5750 (s42) · 8250–9000 (s43)
live EMA + INT4 penalty 2e-4	1	1.22x	1.01x	1.13x	−1 %	0.787	2.5	8250–9000

Uniform INT4 does not pay for itself end-to-end: it wins 24 % on the rollout, loses 21 % on the training side through +28 % tokens, and costs seven reward points. The scheduler keeps the rollout gain, removes the inflation entirely, and lands at 1.14x with reward within one standard error of BF16. The slope-0 reference shows that the downstream term is what drives the late switch; the two seeds settling at 5750 vs 9000 show the online optimum moving with the observed length distribution.

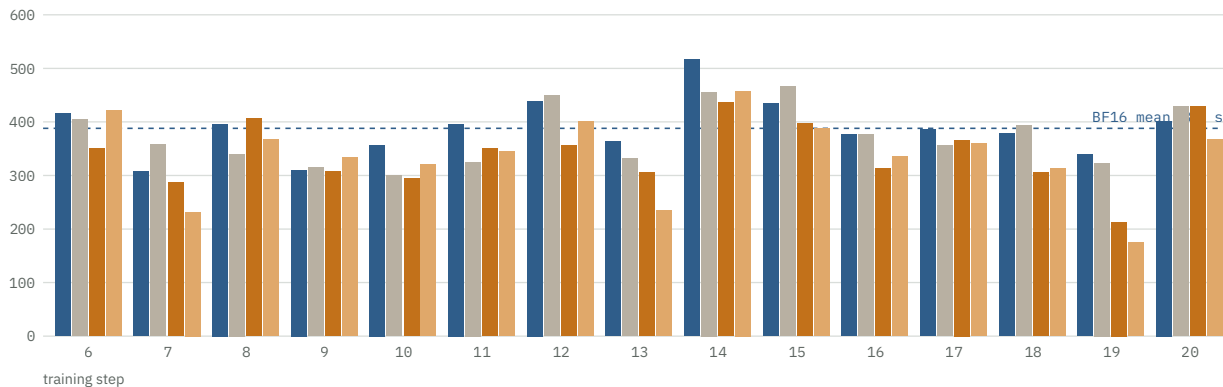
Results: Qwen3.5-4B

ARM	ROLLOUT	DOWNSTREAM	STEP	TOKENS VS BF16	REWARD	SWITCH
BF16, Punica (baseline)	1.00x (188 s)	1.00x (103 s)	1.00x (291 s)	0 %	0.796 ± .020	—
uniform W4, Punica	1.20x	0.84x	1.04x	+19 %	0.798 ± .036	—
live EMA, fitted slope (6.7e-4), ours	1.24x	1.00x	1.14x (254 ± 12 s)	0 %	0.806 ± .028	8250
live EMA + INT4 penalty 2e-4	1.17x	1.02x	1.11x	-1 %	0.812 ± .024	8250-14500

Step by step

■ BF16 Punica (mean of 2 seeds) ■ uniform W4 Punica ■ live EMA, seed 42

■ live EMA, seed 43



The prompt set changes every step, so absolute step time swings 0.7x–1.6x around the mean for every arm (step 14 is the long one, steps 7, 13, 19 the short ones). Against W4, the scheduler wins 1.02–1.54x on 8 of 15 steps and loses 5–10 % on the other 7. The pattern is binary and it is about the tail, not about the switch point.

Where a step's time goes

PHASE OF A 9B LIVE-EMA STEP (339 S)	SECONDS	SHARE	PRECISION
batched decode, tokens 0 → 5750, 32 → ~10 requests alive	~100	30 %	BF16
tail, 1–4 cap-runners decoding alone to 16384	~95	28 %	INT4
training side (old_logprob, ref, update, weight sync)	~145	43 %	—

In 13 of 15 steps at least one response in every arm hits the 16384 cap. Whenever both the scheduler and uniform W4 have a cap-runner, their tails are identical (INT4 at batch 1, ~100 s) and W4 is ahead only by the ~20 s it saves in the batched phase; that is the 0.90–0.95x steps. The scheduler wins the steps where it has no cap-runner or W4 has 2–4x as many long responses, which is the inflation showing up as extra tail. Per step, rollout time correlates 0.55 with the longest response and only 0.33 with total tokens.

Retrospective: would another switch point have helped?

Each step's observed response lengths were replayed through the heatmap (one decode iteration per token, batch = requests still alive, precision by frontier K); the replay reproduces measured rollout time within ~6 %.

K	0	1000	2000	3000	4000	5000	ACTUAL	5750	8000	12000	NEVER
modeled rollout, sum of steps 6–20 (s)	2742	2797	2854	2911	2970	3029		3076	3206	3425	3645

With lengths held fixed, every step's argmin is $K = 0$ and the gain is uniform: about 4 s per 1000 tokens of earlier switching, 22 s per step (11 % of rollout) at most. No step shows a later switch helping, and none shows an earlier switch helping by more than that line. The replay ignores inflation; the slope-0 arm is the empirical check of an early switch: rollout –10 s, but +15 % tokens (+22 s downstream) and reward –0.02, so the step is 12 s slower. The cost model's 5750 is right within the noise (fixed $K = 5000$ measures 348 s vs 339 s).

Why the ratio is lower than the earlier numbers

- **Rollout is 63 % of the BF16 step here.** A 1.27x rollout gain can give at most $1 / (0.37 + 0.63/1.27) = 1.15x$; the measured 1.14x is at that ceiling. The archived headline setting was cap 24576, where the tail is 8k tokens longer and rollout is ~70 % of the step; the same scheduler there projects to roughly 1.25x.
- **The batched phase gains little from INT4.** On the 9B heatmap INT4 is 45 % faster at batch 1 but ~22 % at batch 32, and the scheduler deliberately leaves the batched phase in BF16 because switching there costs 15–28 % tokens. Uniform W4 takes that 22 % and still only reaches 1.02x per step.
- **The tail is a fixed cost.** A cap-runner decodes 16384 tokens alone at ~9.5 ms/token regardless of K; the scheduler already runs all of it in INT4.
- **Earlier "step" numbers were partly modeled.** The rollout-only sweep priced downstream as slope × tokens with no intercept and no training in the loop; with the intercept it read 1.17x, and the measured Megatron value is 1.14x. The direction did not change, the estimate tightened.

The scheduler is doing what the cost model says at this setting. The ceiling is set by the rollout share and by the INT4 gain at batch 16–32, not by where it switches.

Next steps

- **Cap 24576** (the archived setting): BF16 Punica, W4 Punica, live EMA × 2 seeds. Four GPUs, ~3 h. Expected ~1.25x per step for the same scheduler; this is the direct answer to the discrepancy.
- **Cap-runner budget:** rollout-only cap 12k vs 16k on the same prompts to price the last 4k tokens in rollout time and reward. ~14 % of all generated tokens sit in responses that hit the cap.
- **Batched-phase kernels:** only a kernel win at batch 16–32 (fused LoRA / Marlin path) raises the ceiling; that is not a scheduler question.
- **Megatron e2e test** (decision 10 action d) is still open; the gpu-smoke asserts FSDP-only checkpoint paths.

Provenance

- Commits on `ver1-clean` (`rollout-precision-scheduler-clean`): `24508cf3` decision 10 reversed · `acde6425` Megatron driver · `1f96a61c` run report and table script (`examples/precision_scheduler/analysis/mg20_final_table.py`). Not pushed.
- Run dirs: `/data/huanchen/ps_runs/mg20_*` (15 runs), calibration traces `ema{9,4}b_calib_*` ; report `docs/precision_scheduler/RUN_2026-09-16_megatron_mg20.md` .
- Discarded: every FSDP2 training run from 2026-09-15/16 (rep16, e2e full-step, live full-step), deleted at the user's request.